

## Example 1: Die Fairness

Observed counts from 120 rolls: [18, 19, 23, 20, 21, 19].

Expected = 20 each.

$$- \chi^2 = \sum((O - 20)^2/20) \approx 1.8.$$

$$- df = 5, \chi^2_{\{0.05,5\}} = 11.07 \Rightarrow \text{fail to reject; die appears}$$

fair.

## Example 2: Survey Preferences

Categories A/B/C expected 50%/30%/20% ( $p = 0.5, 0.3, 0.2$ ).

Observed counts ( $n = 200$ ): [120, 50, 30].

$$- \text{Expected } E = [100, 60, 40].$$

$$- \chi^2 = ((120-100)^2/100) + ((50-60)^2/60) + ((30-40)^2/40) = 4$$

$$+ 1.67 + 2.5 = 8.17.$$

$$- df = 2, \chi^2_{\{0.05,2\}} = 5.99 \Rightarrow \text{reject } H_0.$$

- Categories A and C deviate most (positive residuals).